

Answer the Quiz 2 Question 1

```
class Foodcart:
    def __init__(self):
        self.list1=[]
        self.cart_name=""
        self.re_food=""
    def addFood(self,*obj):

        if len(self.list1)==0:
            print("Please set the cart name first.")

        for i in obj:
            if i not in self.list1:
                self.list1.append(i)

    def setCartName(self,card_name):
        self.cart_name=card_name
    def printCartDetails(self):
        print(f'Details of Foodpanda')
        print(f'Total Items: {len(self.list1)}')
        for i in self.list1:
            if self.re_food!=i.name:
                print(f'Food info:{i.name}, Food shop: {i.shope_name}, Food price:{i.price}')

    def removeFood(self,food_name):
        self.re_food=food_name
        flag=False
        for i in range(len(self.list1)):
            if self.re_food==self.list1[i].name:
                print(f'{self.list1[i].name} removed from cart!')
                self.list1.pop(i)
                flag=True
                break
        if flag==False:
            print(f'{self.re_food} Fry not found!')

class Food:
    def __init__(self,name,shope_name,price):
```

```
self.name=name
self.shope_name=shope_name
self.price=price
def detail(self):
    print(f'Food info:{self.name}, Food shop: {self.shope_name}, Food
price:{self.price}')
```

Answer the Quiz 2 Question 2

#Quiz 2 Question 2

```
class Team:
```

```
    def __init__(self,country_name):
        self.country_name=country_name
        self.list1=[]
```

```
    def addPlayer(self,*obj):
        for i in obj:
            if i.player_type != "":
                print(f"{i.player_name} added in team")
                self.list1.append(i)
            else:
                print("A player cannot be added without a role.")
```

```
    def showPlayers(self):
        for i in self.list1:
            print(f"{i.player_type}")
            print(f"Player name: {i.player_name} Jersey No: {i.jersey_no}")
```

```
class Player:
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    def __init__(self,*player):
        if len(player)==3:

            self.player_name=player[0]
            self.player_type=player[1]
            self.jersey_no=player[2]
        else:
```

```
            self.player_name=player[0]
            self.player_type=""
            self.jersey_no=player[1]
```

```
    def setRole(self,update_type):
        self.player_type=update_type
```

Answer the MID question 1

```
class SDS:
    def __init__(self,year):
        self.year=year
        self.number_of_cse_student=0
        self.number_of_mns_student=0
        self.cse_student={}
        self.mns_student={}

    def print_info(self):
        if self.number_of_cse_student==0 and self.number_of_mns_student==0:
            print(f'school of data and science {self.year}! only CSE and MNS
department students can be added.")
            print("Number of CSE students:",self.number_of_cse_student)
            print("Number of MNS students:",self.number_of_mns_student)
            print("CSE Students ",self.cse_student)
            print("MNS Students", self.mns_student)

    def addStudent(self,*student_obj):
        for i in student_obj:
            if i.department=="MNS":
                self.number_of_mns_student += 1
                if i.student_id not in self.mns_student:
                    self.mns_student[i.student_id]=[i.student_name,i.cgpa,i.course]
            elif i.department=="CSE":
                self.number_of_cse_student += 1
                if i.student_id not in self.cse_student:
                    self.cse_student[i.student_id]=[i.student_name,i.cgpa,i.course]
            elif i.department=="BBS":
                print(f'{i.department} department student cannot be added to SDS")

class Student:
    def __init__(self,*student_info):
        if len(student_info)==2:
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```
self.student_name=student_info[0]
self.student_id=student_info[1]
self.department="MNS"
self.course=[]
self.cgpa=0
if len(student_info)==3:
    self.student_name = student_info[0]
    self.student_id = student_info[1]
    self.department = student_info[2]
    self.course = []
    self.cgpa = 0
if len(student_info) == 4:
    self.student_name = student_info[0]
    self.student_id = student_info[1]
    self.department = student_info[2]
    self.course = student_info[3]
    self.cgpa = 0
if len(student_info) == 5:
    self.student_name = student_info[0]
    self.student_id = student_info[1]
    self.department = student_info[2]
    self.course = student_info[3]
    self.cgpa = student_info[4]

def print_info(self):
    print("Name",self.student_name)
    print("ID",self.student_id)
    print("Department",self.department)
    print("Course",self.course)
    print("CGPA",self.cgpa)
```

Answer the mid question 2

```
class Dominos:
    def __init__(self,shope_name):
        self.shope_name=shope_name
        self.counter=0
        self.dict_1={}
        self.chicken_dict={}
        self.beef_dict={}
        self.cheese_dict={}
        print(f'{self.shope_name}. The Pizza Delivery Experts.')

    def addPizza(self,*obj):
        for i in obj:
            self.counter+=1
            for j in range(1,len(i.food)):

                # x=i.food[0].split(" ")
                # print(x)
                if "Chicken" in i.food[0]:
                    if i.food[0] not in self.chicken_dict.keys():
                        self.chicken_dict[i.food[0]]=[]
                    self.chicken_dict[i.food[0]].append(i.food[j])
                elif "Beef" in i.food[0]:
                    if i.food[0] not in self.beef_dict.keys():
                        self.beef_dict[i.food[0]]=[]
                    self.beef_dict[i.food[0]].append(i.food[j])
                else:
                    if i.food[0] not in self.cheese_dict.keys():
                        self.cheese_dict[i.food[0]]=[]
                    self.cheese_dict[i.food[0]].append(i.food[j])

    def showMenu(self):
        print(f'Total number of pizzas: {self.counter}')
        print("Chicken pizzas:",len(self.chicken_dict))
        for key,val in self.chicken_dict.items():
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```

    print(f'Name: {key}, Toppings:{val}')
print()
print("Beef pizzas:", len(self.beef_dict))
for key, val in self.beef_dict.items():
    print(f'Name: {key}, Toppings:{val}')
print()
print("cheese pizzas:", len(self.cheese_dict))
for key, val in self.cheese_dict.items():
    print(f'Name: {key}, Toppings:{val}')

```

class Pizza:

```

    def __init__(self,*food):

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        self.list1=[]
        if len(food)<=1:
            self.list1.append(food[0])
            print("A pizza without toppings cannot be created.")
        else:
            for i in food:
                self.list1.append(i)
        self.food = self.list1

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    def setToppings(self,*add_food):

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        print(f"Toppings added to {self.list1[0]} pizza.")
        for i in add_food:
            self.list1.append(i)

```

Answer the mid question 3

```
class Team:
    def __init__(self, country_name, win, language= "Spanish"):
        self.country_name=country_name
        self.win=win
        self.language=language
        self.player_list ="No players assigned yet"
        self.cost = 0
        self.forward_list=[]    #[Messi, Maria]
        self.Defender_list=[]
    def calculateCost(self):
        print("No cost yet as there is no players")
    def printDetail(self):
        print("Team Name:",self.country_name)
        print("World Cup Win:",self.win)
        print("Language:",self.language)
        if len(self.forward_list)!=0:
            print("Forwards:")
            for i in self.forward_list:
                print(i)
            print("Defenders:")
            for i in self.Defender_list:
                print(i)
    def addPlayer(self,player_obj):
        if type(self.player_list) !=type([]):
            self.player_list=[]

self.player_list.append((player_obj.player_name,player_obj.player_position))
    if player_obj.player_position=="Forward":
        self.forward_list.append(player_obj.player_name)
        self.cost+=500
    elif player_obj.player_position=="Defender":
        self.Defender_list.append(player_obj.player_name)
        self.cost+=450
    def calculateCost(self):
        if len(self.forward_list)!=0 or len(self.Defender_list)!=0:
```



```
        print("Total cost:",self.cost)
class Player:
    def __init__(self,player_name,player_position="Forward"):
        self.player_name=player_name
        self.player_position=player_position
```

Answer the mid question 4

```
class Vehicle:
    def __init__(self,name,type,type_name,chart):
        self.name=name
        self.type=type
        self.type_name=type_name

        self.counter=0
        self.dict_1={}

    def addPassenger(self,*obj):
        x=self.name.split("-")

        for i in obj:
            if x[0]==i.location:
                self.counter+=1
                print(f'{i.p_name} has taken the {self.type}')
                if i.p_name not in self.dict_1.keys():
                    self.dict_1[i.p_name]=x[0]
            else:
                print(f'This is an Inter-City {self.type}. {i.p_name} cannot take the {self.type}.')

    def status(self):
        print(f'Details of {self.name} {self.type} of {self.type_name}')
        print(f'Total passenger: {self.counter}')
        for key,val in self.dict_1.items():
            if self.type_name in fareChart.values():
                print(f'Passenger name : {key}, Destination: {val}, Ticket Price: '
                    f'{fareChart[self.type_name]}')
            else:
                print(f'Passenger name : {key}, Destination: {val}, Ticket Price: '
                    f'{90}')
```

```
class Passenger:
    def __init__(self,p_name,area=""):
        self.p_name=p_name
        self.location=area
        if self.location=="":
            print("Destination cannot be empty!")

    def calculateFare(self,obj1):
        print(f'Calculating fare for {self.p_name}')

    def setDestination(self,add_area):
        print(f'Destination of {self.p_name} is updated to {add_area}')
```

Answer the mid question 5

```
class Passenger:
    def __init__(self,*args):
        if len(args)==3:
            self.name=args[0]
            self.type=args[1]
            self.amount=args[2]
        else:
            self.name=args[0]
            self.type=""
            self.amount=args[1]
        if self.type !="":
            print(f'{self.name} is waiting for {self.type.split(" ")[0]} having {self.amount}
taka in pocket')
        else:
            print(f'{self.name} has {self.amount} taka in pocket. please buy a ticket')

    def buyTicket(self,add_type):
        self.type=add_type

class TicketCounter:
    def __init__(self):
        print(f'This is a general bus and train ticket counter')
        self.bus=0
        self.train=0
        self.bus_dict={}
        self.train_dict={}

    def servePassenger(self,*obj):
        for i in range(len(obj)-1):
            if obj[i].type!="":
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```
if obj[i].type=="Bus Ticket":
    self.bus+=1
    if obj[i].name not in self.bus_dict.keys():
        self.bus_dict[obj[i].name]=obj[i].amount-obj[-1]["Bus"]
else:
    self.train+=1
    if obj[i].name not in self.train_dict.keys():
        self.train_dict[obj[i].name]=obj[i].amount-obj[-1]["Train"]
    print(f'{obj[i].name} has bought {obj[i].type} ticket ')
else:
    print(f'{obj[i].name} please buy a ticket first')
```

```
def showAllPassenger(self):
    print(f'Total passengers: {self.bus+self.train}')
    print(f'Bus passengers: {self.bus}')
    for key,val in self.bus_dict.items():
        print(f'Passenger Name: {key} Remaining Balance: {val}')
    print(f'Train passengers: {self.train}')
    for key,val in self.train_dict.items():
        print(f'Passenger Name: {key} Remaining Balance: {val}')
```